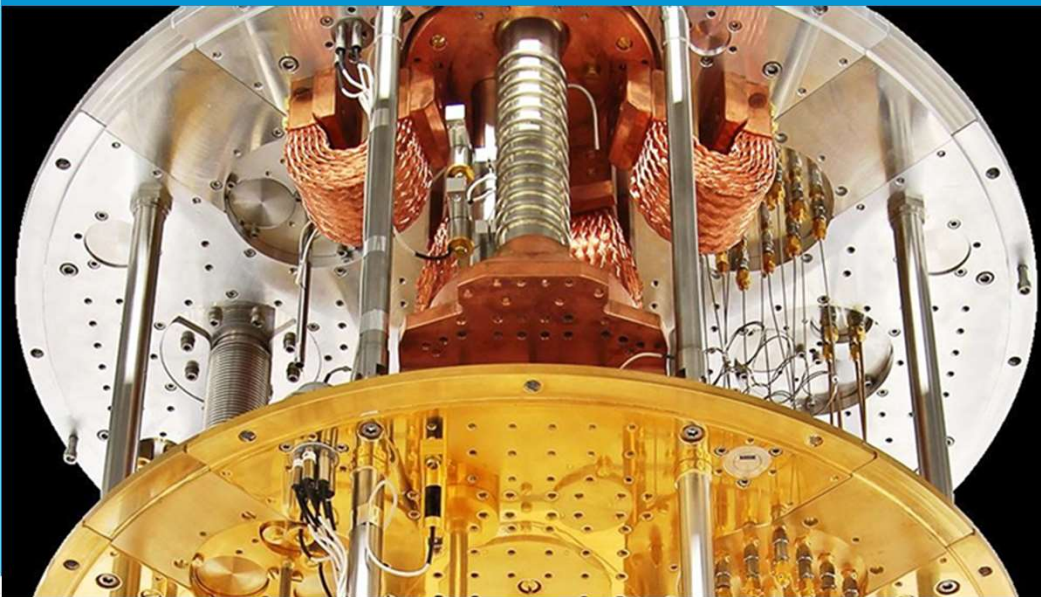


Quantum Inspire Interactive Demo

HAROLD MEERWALDT – SOFTWARE ENGINEER – H.B.MEERWALDT@TUDELFT.NL



QuTech is a collaboration between **TU Delft** **TNO**

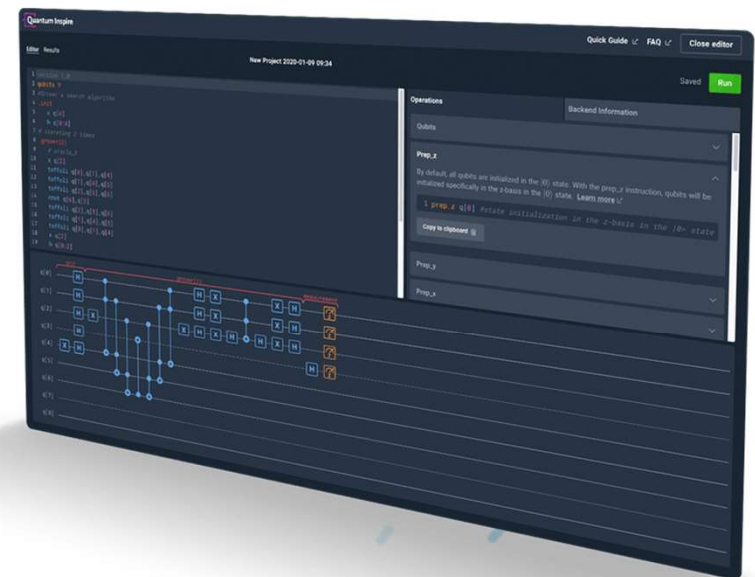
Quantum Inspire



A prototype **full-stack quantum computer**, focused on training, education, and the development of applications

- Europe's first quantum computer in the cloud
- Provides access to:
 - a processor of spin qubits (world's first)
 - a processor of superconducting (transmon) qubits
 - quantum emulators (also connected to a supercomputer)
- Web frontend and Python SDK
- Knowledge base

→ quantum-inspire.com



Grover's search algorithm

Developed by Lov Grover in 1996

Unstructured search algorithm

Suppose you have a function f operating on $x : f(x)$

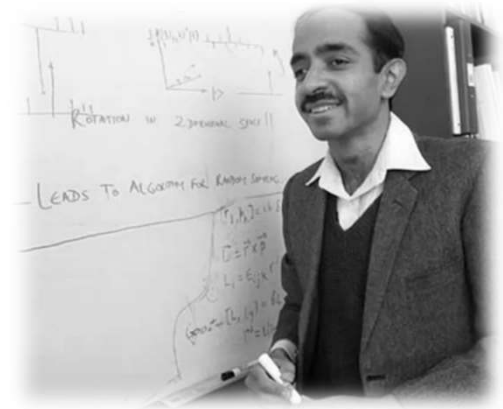
$$f(x) = \begin{cases} 1 & \text{if } x = x_M \\ 0 & \text{if } x \neq x_M \end{cases} \quad \text{with } x_M \text{ the input we are searching for}$$

Classical algorithms require $O(N)$ calls to $f(x)$

Grover's algorithm requires $O(\sqrt{N})$ calls to $f(x)$

Thus yielding a quadratic speed up

Use cases: Unstructured search, satisfiability, ...



<https://www.quantum-inspire.com/kbase/grover-algorithm/> 3

Classical bits and quantum bits

A classical bit:

is in a state 0 or 1

if state 0,
measurement $\rightarrow$ 0
afterwards: state 0

if state 1
measurement $\rightarrow$ 1
afterwards: state 1

state = measured value

A quantum bit (qubit):

is in a state: $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$

- $\alpha_0, \alpha_1 \in \mathbb{C} \rightarrow$ complex amplitudes
- if $\alpha_0, \alpha_1 \neq 0 \rightarrow$ superposition

measurement $\rightarrow$ 0 or 1

- $P_0 = |\alpha_0|^2, P_1 = |\alpha_1|^2 \rightarrow |\alpha_0|^2 + |\alpha_1|^2 = 1$
- probabilities depend on magnitudes

if measurement 0:
afterwards: $|\psi\rangle = |0\rangle$

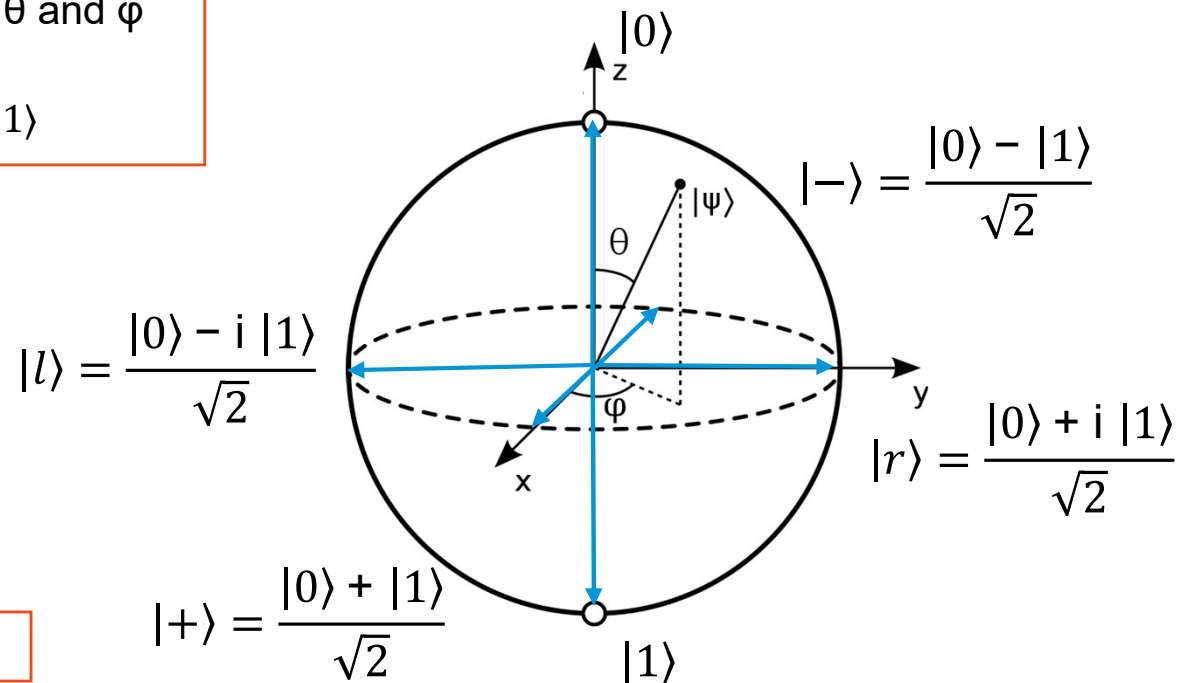
if measurement 1:
afterwards: $|\psi\rangle = |1\rangle$

state $\neq$ measured value
measurement changes state

Bloch sphere

A qubit state $|\psi\rangle$ can be described as a vector on a sphere (Bloch sphere) using polar angles θ and φ

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + \exp(i\varphi) \sin\left(\frac{\theta}{2}\right) |1\rangle$$

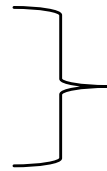


Single-qubit operations / gates as rotations

X gate

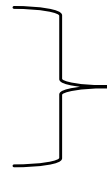
The X gate rotates 180° around the X-axis

$$\begin{aligned} X|0\rangle &= |1\rangle \\ X|1\rangle &= |0\rangle \end{aligned}$$

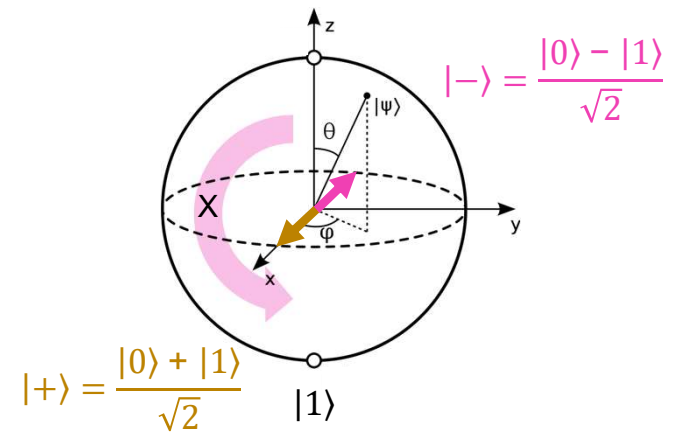
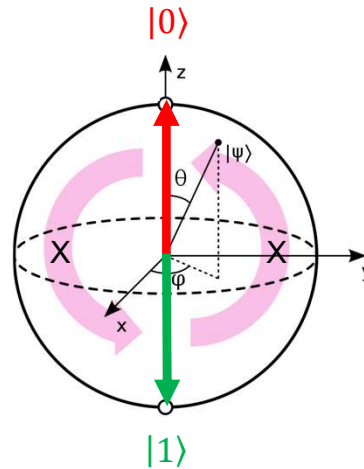


NOT

$$\begin{aligned} X|+\rangle &= |+\rangle \\ X|-\rangle &= -|-\rangle \end{aligned}$$



possible
amplitude flip



H gate

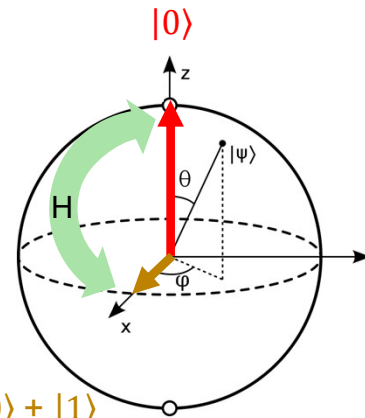
The H gate (Hadamard gate) rotates:

- 90° around the Y-axis
- then 180° around the X-axis

$$H|0\rangle = |+\rangle$$

$$H|1\rangle = |-\rangle$$

make superposition

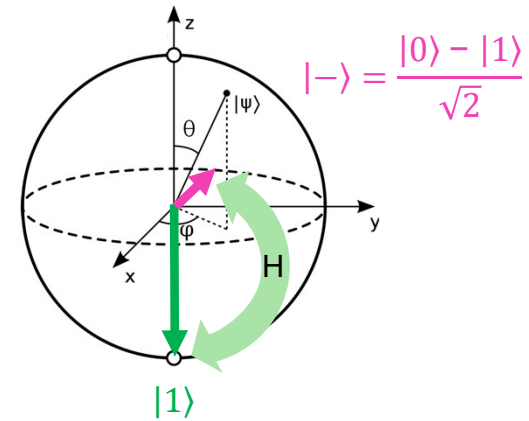


$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|+\rangle = |0\rangle$$

$$H|-\rangle = |1\rangle$$

undo superposition



$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$H^2|\psi\rangle = |\psi\rangle$$

Let's try this out ...



Multiple qubits

Product state:

$$|\psi\rangle = (\underbrace{\alpha_0|0\rangle + \alpha_1|1\rangle}_{\text{qubit 1}}) \otimes (\underbrace{\beta_0|0\rangle + \beta_1|1\rangle}_{\text{qubit 2}})$$

$$= \alpha_0\beta_0|00\rangle + \alpha_1\beta_0|01\rangle + \alpha_0\beta_1|10\rangle + \alpha_1\beta_1|11\rangle$$

qubit 1 is the least significant bit

An entangled state, e.g. a Bell state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

cannot be written as a product state

CNOT gate

The CNOT gate (controlled-NOT or CX gate) applies:

- an X gate on the target qubit
- if the control qubit is in the $|1\rangle$ state
- otherwise do nothing

For control = 1st qubit, target = 2nd qubit:

$$\text{CNOT}|00\rangle = |00\rangle$$

$$\text{CNOT}|10\rangle = |10\rangle$$

} no X applied

$$\text{CNOT}|01\rangle = |11\rangle$$

$$\text{CNOT}|11\rangle = |01\rangle$$

} X applied on target qubit

Let's make a Bell state ...



Toffoli gate

The Toffoli gate (or CCNOT or CCX) applies:

- an X gate on the target qubit
- if its two control qubits are in the $|1\rangle$ state
- otherwise do nothing

For control = 1st and 2nd qubit, target = 3rd qubit:

$$\begin{array}{l} \textit{Toffoli}|000\rangle = |000\rangle \\ \textit{Toffoli}|010\rangle = |010\rangle \\ \textit{Toffoli}|001\rangle = |001\rangle \end{array} \left. \vphantom{\begin{array}{l} \textit{Toffoli}|000\rangle = |000\rangle \\ \textit{Toffoli}|010\rangle = |010\rangle \\ \textit{Toffoli}|001\rangle = |001\rangle \end{array}} \right\} \text{no X applied}$$

$$\begin{array}{l} \textit{Toffoli}|011\rangle = |111\rangle \\ \textit{Toffoli}|111\rangle = |011\rangle \end{array} \left. \vphantom{\begin{array}{l} \textit{Toffoli}|011\rangle = |111\rangle \\ \textit{Toffoli}|111\rangle = |011\rangle \end{array}} \right\} \text{X applied on target qubit}$$

Small recap

Single-qubit gates:

- X $\rightarrow$ rotate 180° around the X-axis
- H $\rightarrow$ make/undo superposition

Multi-qubit gates:

- CNOT $\rightarrow$ apply X on target if control is in the $|1\rangle$ state
- Toffoli $\rightarrow$ apply X on target if both controls are in the $|1\rangle$ state

Grover's search algorithm

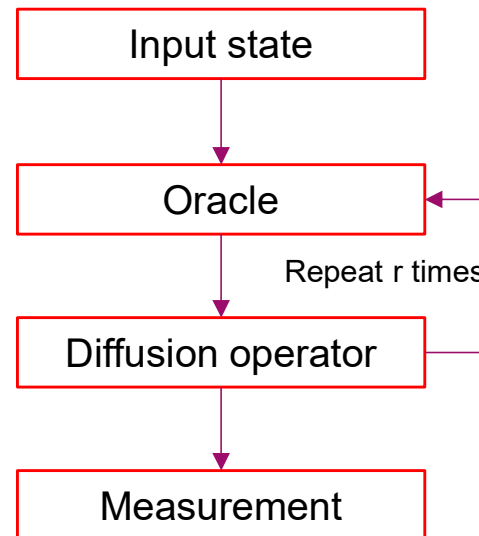
Requires four components

1) Oracle function $f(x)$

2) Superposition input state

3) Diffusion operator (amplification)

4) Measurement



Oracle

An operator that:

- marks the solution to the search problem
- by flipping the amplitude of the solution

If

$$|\psi\rangle = \alpha_0\beta_0|00\rangle + \alpha_1\beta_0|01\rangle + \alpha_0\beta_1|10\rangle + \alpha_1\beta_1|11\rangle$$

and

$|11\rangle$ is the solution to the search problem,

then

$$\text{Oracle}|\psi\rangle = \alpha_0\beta_0|00\rangle + \alpha_1\beta_0|01\rangle + \alpha_0\beta_1|10\rangle - \alpha_1\beta_1|11\rangle$$



amplitude flipped

Making an oracle

We can make an oracle with:

- a Toffoli gate
- Hadamard gates

remembering

$$H|0\rangle = |+\rangle$$

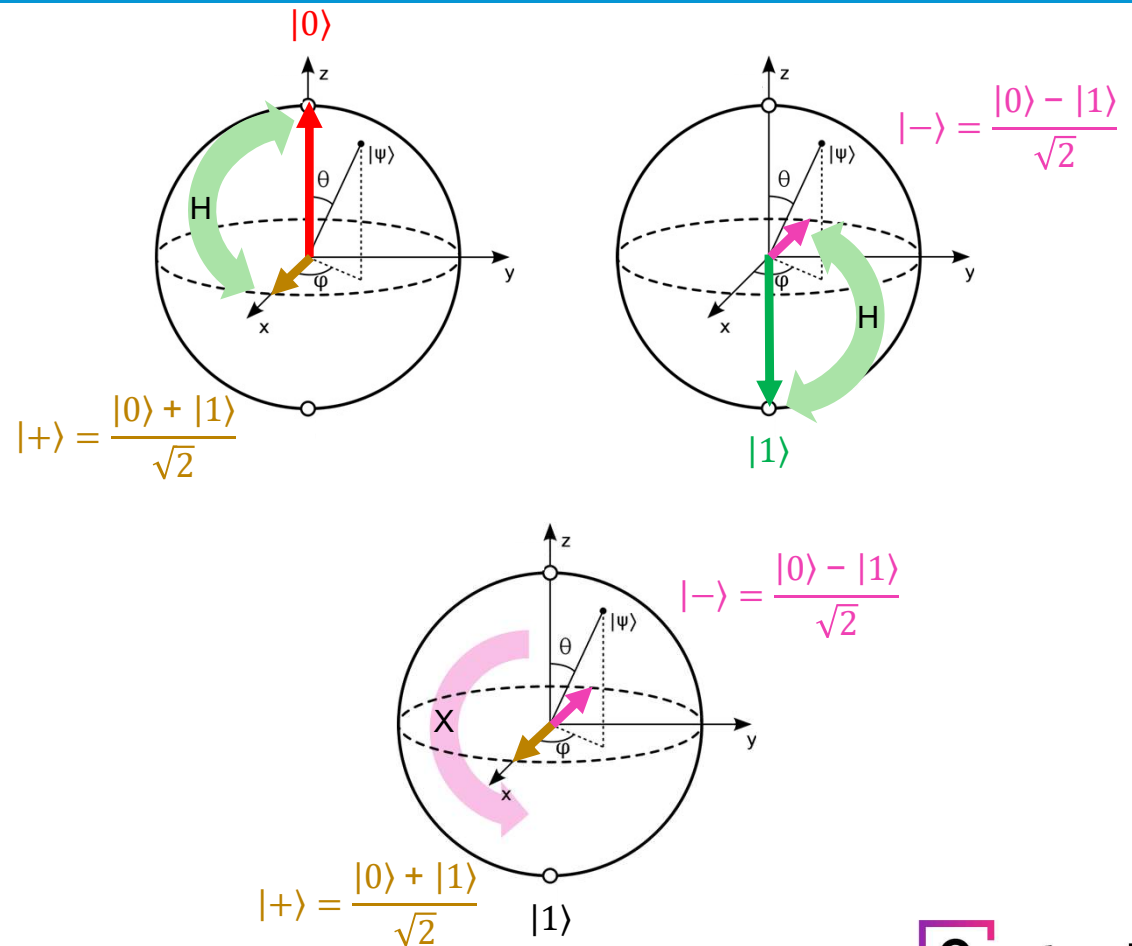
$$H|1\rangle = |-\rangle$$

and

$$X|+\rangle = |+\rangle$$

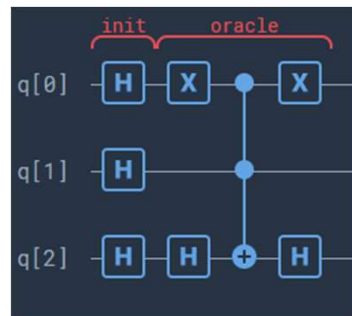
$$X|-\rangle = -|-\rangle$$

Let's make one with $|110\rangle$ as solution

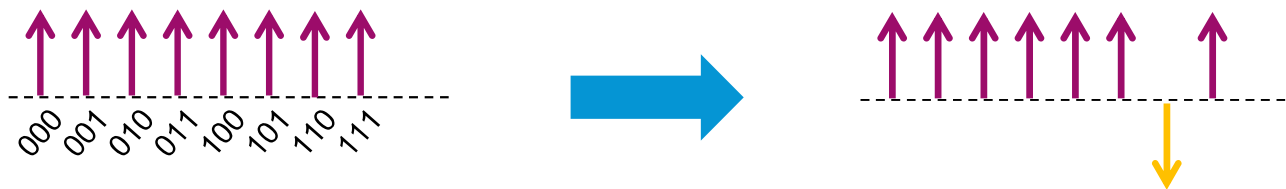


Adding a superposition input state

Now we need to input all possible states into the oracle.



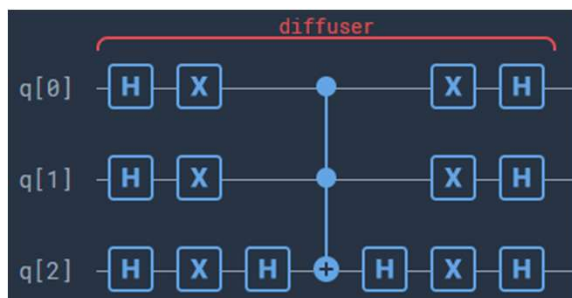
A superposition input state followed by the oracle flips the amplitude of the solution state, here $|110\rangle = \text{"6"}$



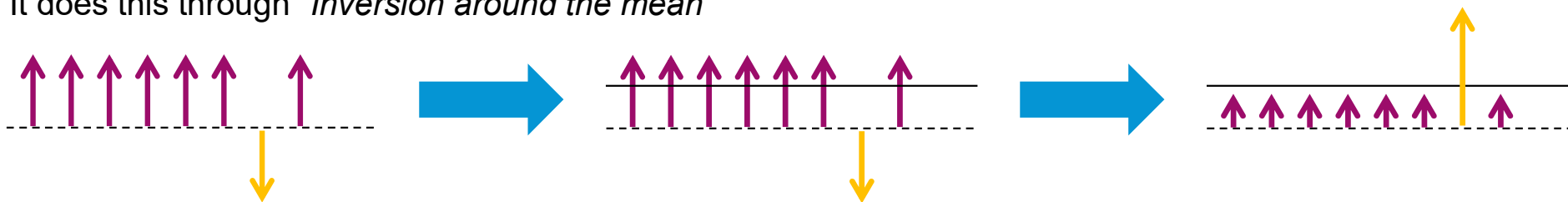
But we cannot measure amplitude directly!

Transforming amplitude differences into magnitude differences

The diffusion operator amplifies amplitude differences



It does this through “*Inversion around the mean*”



Grover iterations

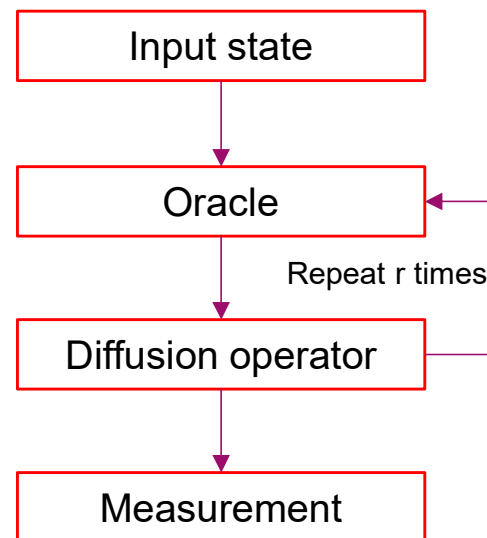
Requires four components

1) Oracle function $f(x)$

2) Superposition input state

3) Diffusion operator (amplification)

4) Measurement



$$r = \frac{\pi}{4} \sqrt{\frac{N}{M}} = \frac{\pi}{4} \sqrt{\frac{2^n}{M}}$$

N , number of states

n , number of qubits

M , number of solutions

$$n = 3, M = 1 \rightarrow r \approx 2.2$$

Boolean satisfiability problem

The solution was already programmed into the Oracle.

Can we satisfy a Boolean problem?

E.g. for which values is:

$(q[0] \text{ OR } q[1]) \text{ AND NOT } q[2]$

True?

- 001
- 010
- 011

→ $M = 3$

Boolean operators as quantum gates

NOT $\rightarrow$ X

AND $\rightarrow$ Toffoli

OR $\rightarrow$ build up from NOT (X) and AND (Toffoli)

We need scratch qubits (aka ancilla qubits) for the AND and the OR

Input into Oracle

Let's build the Boolean circuit as input for the Oracle

$(q[0] \text{ OR } q[1]) \text{ AND NOT } q[2]$

We need $r = 1.8$ iterations (with $n=4$ and $M=3$)

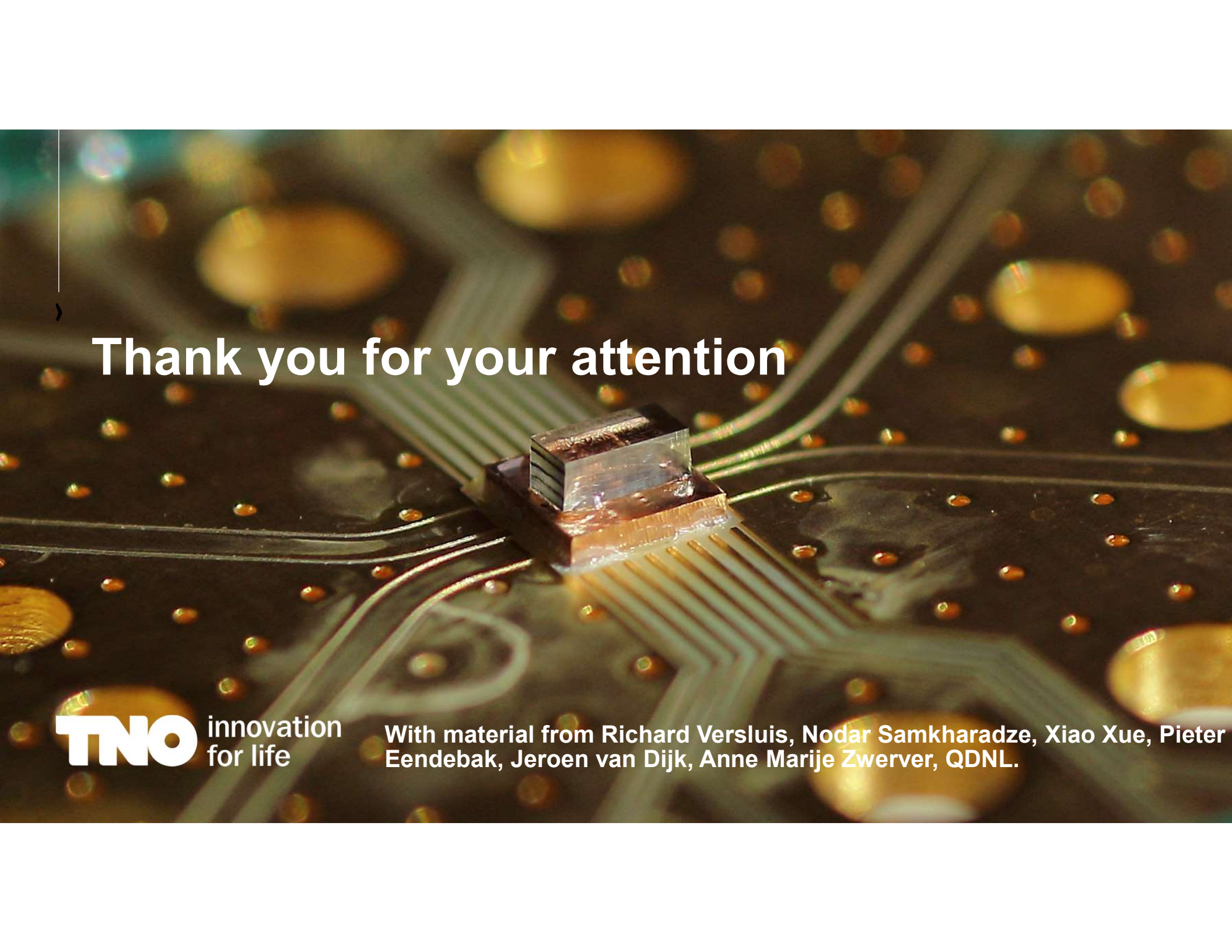
Where to go from here?

Have a look at the knowledge base,
and try out some algorithms yourself

<https://www.quantum-inspire.com/kbase/>

What's next for Quantum Inspire?

- Connect new systems:
 - 4 spin qubits (2024)
 - Rydberg atoms
- Hybrid quantum-classical algorithms (2024)
- Quantum puzzle game (2024)

A microscopic view of a microchip, showing a central component with a transparent, rectangular, layered structure. This component is connected to a network of fine, parallel metal lines (traces) that radiate outwards across the dark surface of the chip. Numerous small, circular gold-colored pads are visible on the surface, some of which are connected to the traces. The lighting creates a warm, golden glow, highlighting the intricate details of the microelectronics.

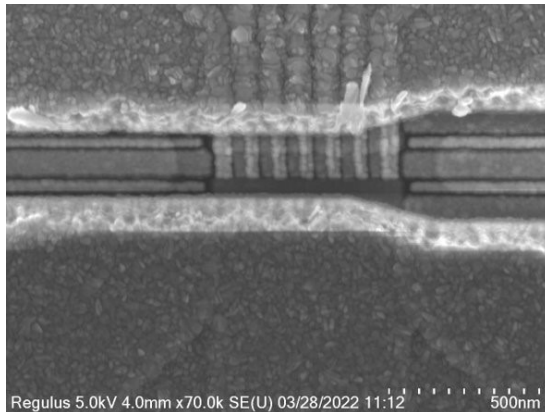
Thank you for your attention

TNO innovation
for life

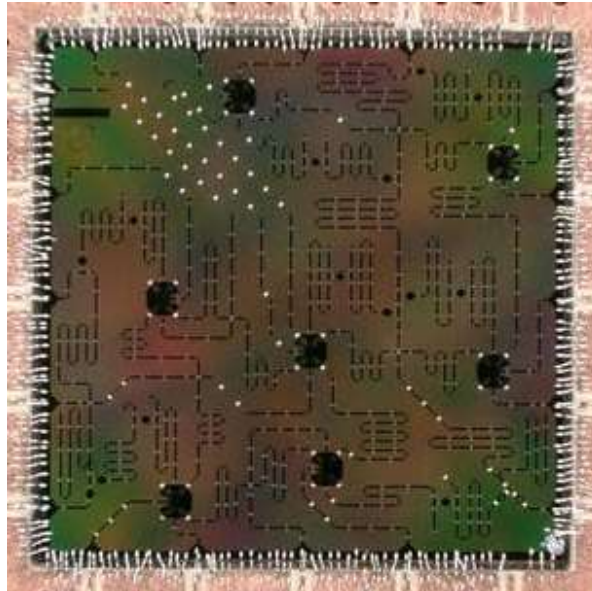
With material from Richard Versluis, Nodar Samkharadze, Xiao Xue, Pieter Eendebak, Jeroen van Dijk, Anne Marije Zwerver, QDNL.

Qubit types at QuTech

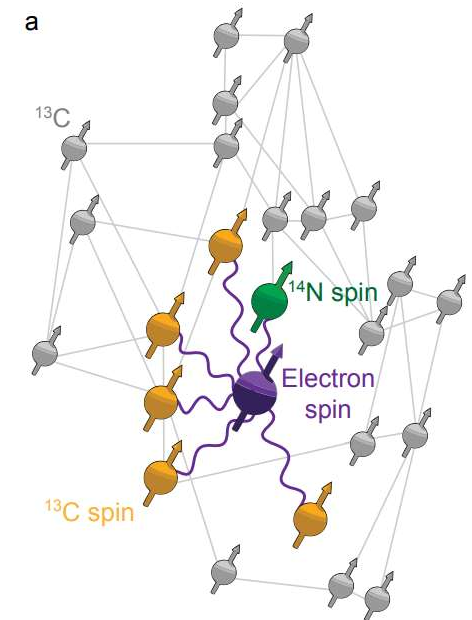
Electron (or hole) spin qubits
in semiconductor quantum dots



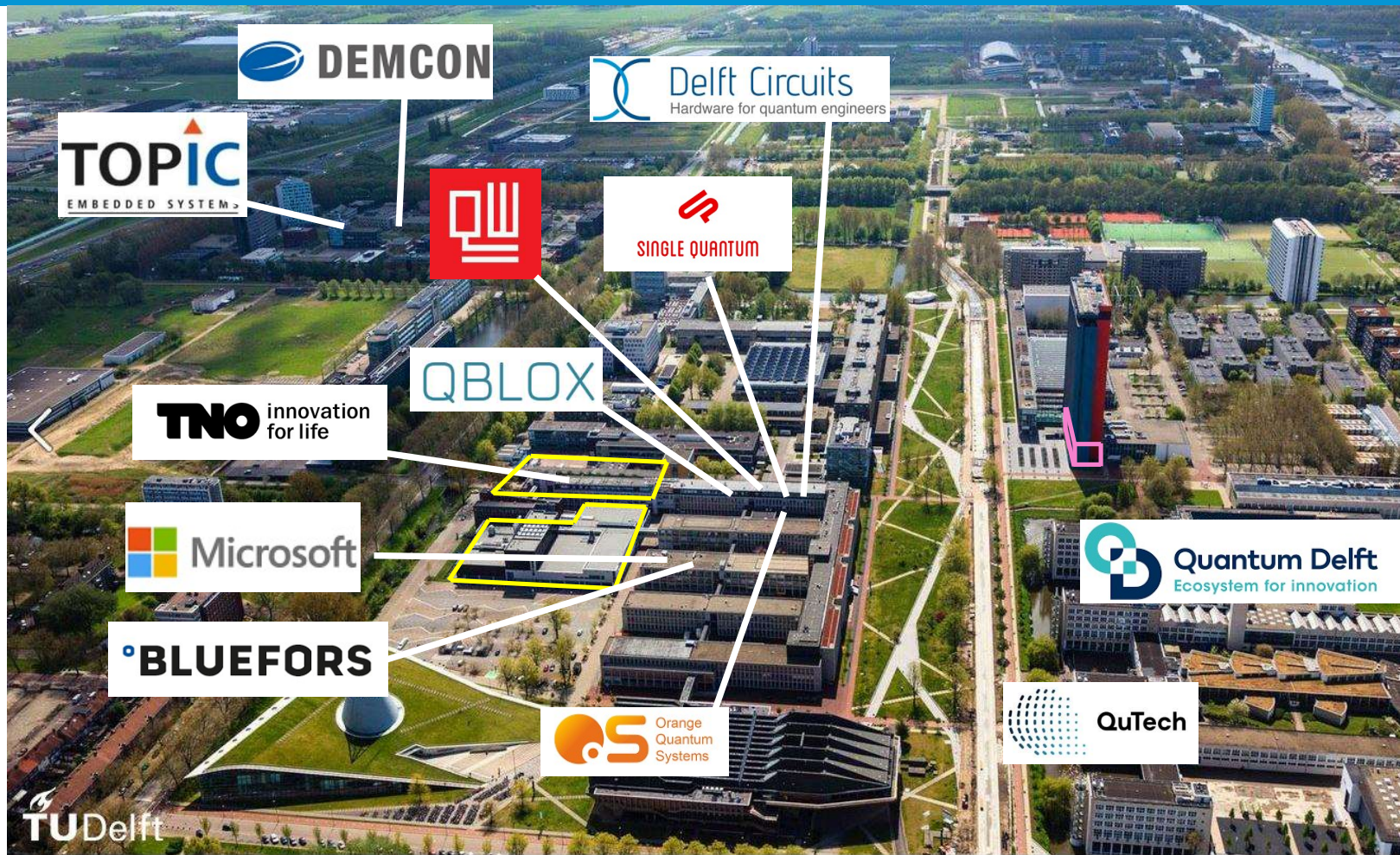
Superconducting qubits
Transmon, fluxonium etc



Spins of impurities in
a diamond crystal



The Delft ecosystem



Grover 1 iteration → circuit

```
version 1.0

qubits 3

.init
H q[0:2]

.oracle
{X q[0] | H q[2]}
Toffoli q[0], q[1], q[2]
{X q[0] | H q[2]}

.diffuser
H q[0:2]
X q[0:2]
H q[2]
Toffoli q[0], q[1], q[2]
H q[2]
X q[0:2]
H q[0:2]
```

Grover 1 iteration → results

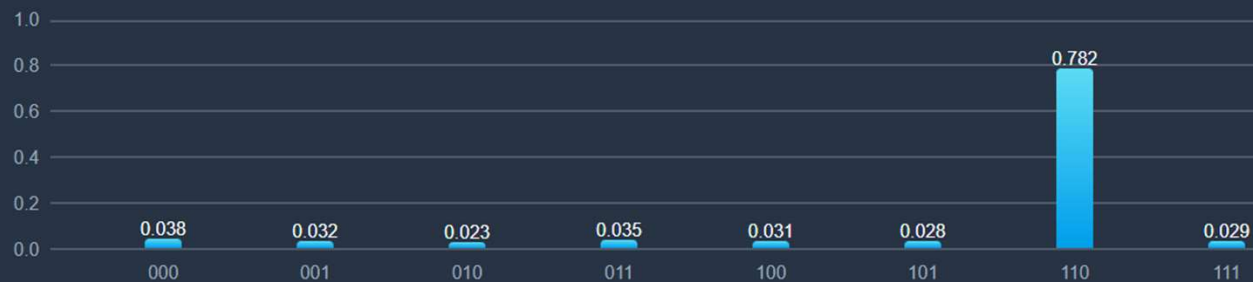
New Experiment 5/3/2024 - 11:36:44 AM | Optimized

Re-run

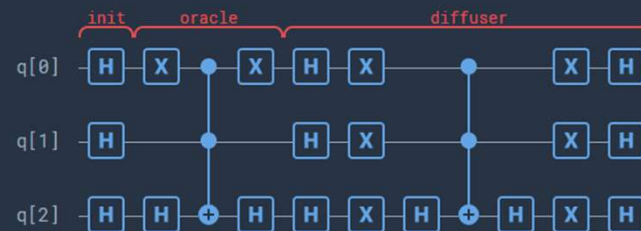
Device: QX single-node simulator
Number of shots: 1024
Execution time: 0.078s

Histogram data:
Raw data:

[CSV](#) [JSON](#)
[CSV](#) [JSON](#)



```
8 .oracle
9 {X q[0] | H q[2]}
10 Toffoli q[0], q[1], q[2]
11 {X q[0] | H q[2]}
12
13 .diffuser
14 H q[0:2]
15 X q[0:2]
16 H q[2]
17 Toffoli q[0], q[1], q[2]
```



Export

Grover 2 iterations → circuit

```
version 1.0

qubits 3

.init
H q[0:2]

.iteration(2)
{X q[0] | H q[2]}
Toffoli q[0], q[1], q[2]
{X q[0] | H q[2]}

H q[0:2]
X q[0:2]
H q[2]
Toffoli q[0], q[1], q[2]
H q[2]
X q[0:2]
H q[0:2]
```

Grover 2 iterations → results

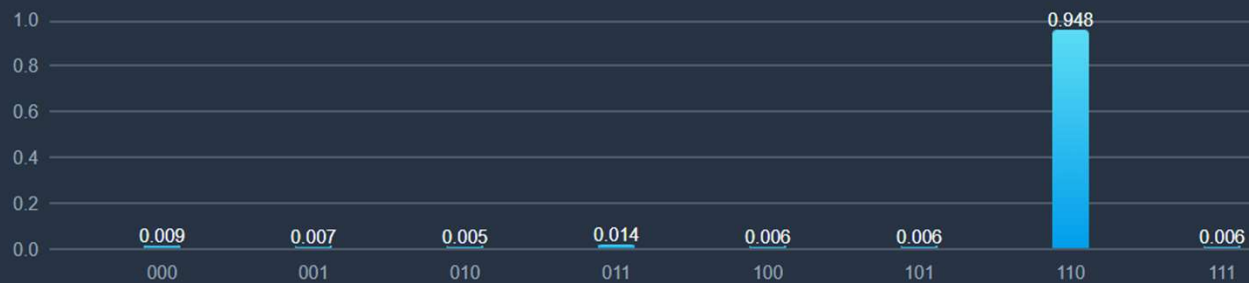
New Experiment 5/3/2024 - 11:37:54 AM | Optimized

Re-run

Device: QX single-node simulator
Number of shots: 1024
Execution time: 0.080s

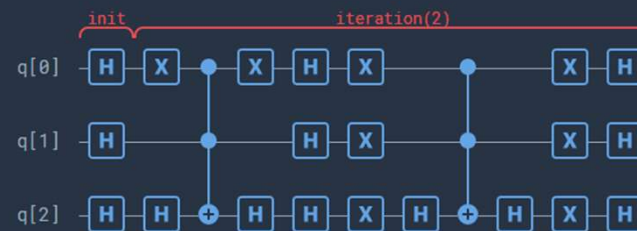
Histogram data:
Raw data:

[CSV](#) [JSON](#)
[CSV](#) [JSON](#)



```
1 version 1.0
2
3 qubits 3
4
5 .init
6 H q[0:2]
7
8 .iteration(2)
9 {X q[0] | H q[2]}
10 Toffoli q[0], q[1], q[2]
```

Export



Grover 3 iterations

```
version 1.0

qubits 3

.init
H q[0:2]

.iteration(3)
{X q[0] | H q[2]}
Toffoli q[0], q[1], q[2]
{X q[0] | H q[2]}

H q[0:2]
X q[0:2]
H q[2]
Toffoli q[0], q[1], q[2]
H q[2]
X q[0:2]
H q[0:2]
```

Grover 3 iterations

New Experiment 5/3/2024 - 11:38:14 AM | Optimized

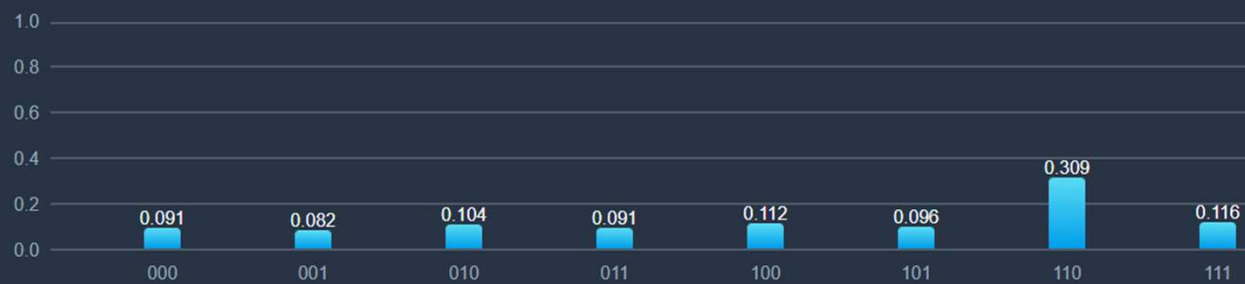
Re-run

Device: QX single-node simulator
Number of shots: 1024
Execution time: 0.067s

Histogram data:
Raw data:

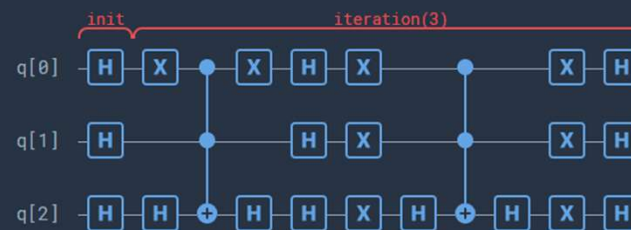
[CSV](#) [JSON](#)

[CSV](#) [JSON](#)



```
1 version 1.0
2
3 qubits 3
4
5 .init
6 H q[0:2]
7
8 .iteration(3)
9 {X q[0] | H q[2]}
10 Toffoli q[0], q[1], q[2]
```

Export



Boolean satisfiability: $(q[0] \text{ OR } q[1]) \text{ AND NOT } q[2] \rightarrow \text{circuit}$

```
version 1.0
```

```
qubits 5
```

```
.init
```

```
H q[0:2, 4]
```

```
.oracle
```

```
X q[0:1]
```

```
Toffoli q[0], q[1], q[3]
```

```
X q[0:1,3]
```

```
{X q[2] | H q[4]}
```

```
Toffoli q[2], q[3], q[4]
```

```
{X q[2] | H q[4]}
```

```
X q[0:1,3]
```

```
Toffoli q[0], q[1], q[3]
```

```
X q[0:1]
```

```
.diffuser
```

```
H q[0:2, 4]
```

```
X q[0:2, 4]
```

```
H q[4]
```

```
Toffoli q[0], q[1], q[3]
```

```
Toffoli q[2], q[3], q[4]
```

```
Toffoli q[0], q[1], q[3]
```

```
H q[4]
```

```
X q[0:2,4]
```

```
H q[0:2,4]
```

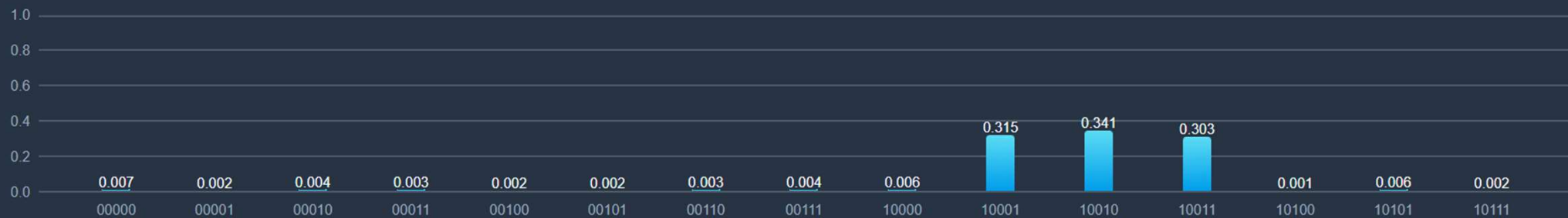
Boolean satisfiability: $(q[0] \text{ OR } q[1]) \text{ AND NOT } q[2] \rightarrow \text{results}$

New Experiment 5/3/2024 - 11:45:58 AM | Optimized

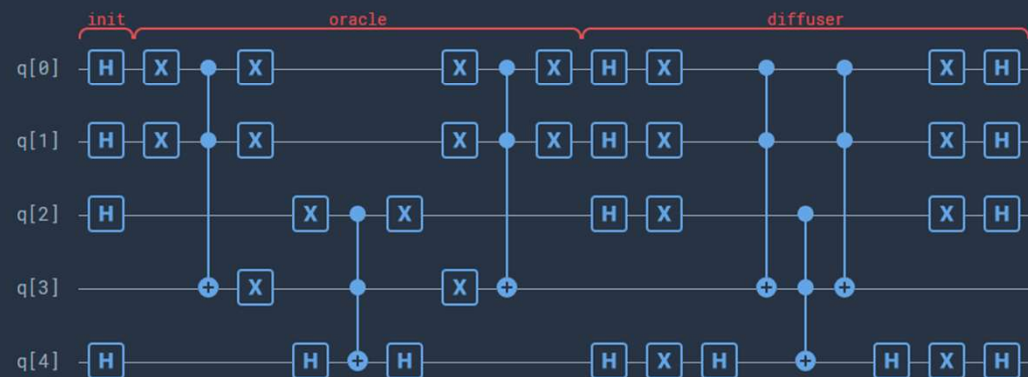
Re-run

Device: QX single-node simulator
Number of shots: 1024
Execution time: 0.111s

Histogram data: [CSV](#) [JSON](#)
Raw data: [CSV](#) [JSON](#)



```
1 version 1.0
2
3 qubits 5
4
5 .init
6 H q[0:2, 4]
7
8 .oracle
9 X q[0:1]
10 Toffoli q[0], q[1], q[3]
11 X q[0:1,3]
12
13 {X q[2] | H q[4]}
14 Toffoli q[2], q[3], q[4]
15 {X q[2] | H q[4]}
```



Export